



Lesson 3: Substance Effects Chart

Analyzing short-term and long-term effects.

 In this activity, you will analyze the immediate and long-term health effects of various substances. By exploring these impacts, you will better understand how substances influence the human body over time.

Setting the Stage

Understanding how different substances interact with the human body is a **vital health literacy skill**. Substances can alter brain chemistry and physical health, leading to varying consequences depending on the duration of use. Learning to distinguish between **short-term** and **long-term** effects helps us make informed decisions about our own health and wellbeing.

What is a substance?	A substance is any chemical or biological agent that alters the physical or mental state of a person. This includes everything from medicinal drugs to recreational substances.
What is a short-term effect?	A short-term effect is a temporary change that happens almost immediately after using a substance. These often include changes in heart rate, mood, or perception.
What is a long-term effect?	A long-term effect is a health change that develops over a prolonged period of repeated substance use. These can result in permanent damage to organs or chronic health conditions.

Read the Facts

Read the following information regarding substance use carefully. As you read, ask yourself: **How does the duration of substance use change its impact on the body?**

Substances affect the body in complex ways. In the short term, many substances disrupt normal brain function, leading to impaired judgment, altered coordination, or heightened risk-taking behavior. For instance, stimulants can cause a rapid increase in energy and heart rate, while depressants like alcohol slow down central nervous system processes.

The risk grows significantly when usage becomes chronic. Long-term effects are often subtle at first but can lead to severe health crises. Prolonged nicotine exposure is linked to respiratory failure and cardiovascular disease, while chronic marijuana use can impact cognitive development in adolescents. Prescription misuse remains a high-risk behavior that can lead to dependency and organ failure over time.

Ultimately, understanding these patterns is key to health. By knowing how substances change your body from the first use to years later, you are better equipped to protect your future.

Building Connections

Connect the information you just read to your own knowledge by filling out this chart.

Key Concept	Your Understanding	Why is this important?
Short-term impact	Immediate changes to brain or body function after use.	It helps us recognize immediate danger signals.
Long-term health consequences	Write your answer here	Write your answer here
Dependency and health risks		
+ Add a row or column as needed		

Try It Out

Use what you have learned to solve the challenges below.

Scenario Question	Response
If a peer believes that vaping is harmless because it is short-term, how would you respond using evidence?	Write your answer here
Explain why chronic prescription misuse is often considered more dangerous than a one-time error.	
Predict one long-term benefit of choosing to avoid substance use during adolescence.	



Show What You Learned

Create a **Digital Visual Infographic** to share what you have learned. Use Magic Media or Elements to help visualise or record your idea. Follow these steps:

- Select one substance from the list.
- Create a split-screen design showing Short-Term vs. Long-Term effects.
- Use icons or images to represent the health risks.
- Write a short summary explaining why prevention is the best health strategy.

[Double click to design here]



Reflect on Your Learning

Think back on the work you have completed regarding substance health effects.

How can differentiating between short-term and long-term effects change the way a person views substance use?

[Write your answer here.]

What was the most surprising piece of information you learned about the long-term impact of substances?

How will this knowledge influence the health choices you make in the future?